

# HEALTH INSURANCE PREMIUMS - FACTORS AND PREDICTIONS

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## Contents

|                                                                         |           |
|-------------------------------------------------------------------------|-----------|
| <b>SECTION 1 - SUMMARY</b>                                              | <b>2</b>  |
| <b>SECTION 2 - INTRODUCTION</b>                                         | <b>2</b>  |
| <b>SECTION 3 - EXPLORATORY DATA ANALYSIS</b>                            | <b>3</b>  |
| <b>SECTION 4 - DATA PREPARATION</b>                                     | <b>5</b>  |
| <b>SECTION 5 - FIVE MODELS</b>                                          | <b>7</b>  |
| SUB-SECTION 5.1 - MODEL 1 - MULTIPLE LINEAR REGRESSION MODEL . . . . .  | 7         |
| SUB-SECTION 5.2 - MODEL 2 - STOCHASTIC GRADIENT DESCENT MODEL . . . . . | 10        |
| SUB-SECTION 5.3 - MODEL 3 - GENERALIZED ADDITIVE MODEL (GAM) . . . . .  | 13        |
| SUB-SECTION 5.4 - MODEL 4 - HIGH DIMENSIONAL MODEL . . . . .            | 17        |
| SUB-SECTION 5.5 - MODEL 5 - BOOSTED REGRESSION TREES MODEL . . . . .    | 21        |
| SUB-SECTION 5.6 - MODELS SUMMARY . . . . .                              | 24        |
| <b>SECTION 6 - LIMITATIONS AND IMPROVEMENTS</b>                         | <b>24</b> |
| <b>SECTION 7 - FURTHER WORK DIRECTIONS</b>                              | <b>25</b> |
| <b>SECTION 8 - CONCLUSION</b>                                           | <b>26</b> |
| <b>SECTION 9 - REFERENCES</b>                                           | <b>26</b> |

## SECTION 1 - SUMMARY

The study focuses on analyzing socio-demographic characteristics which determine policy holders' insurance premiums and predicting those premiums. We develop five statistical models using various Machine Learning techniques. We discover that in general, being older, having higher BMI and more dependents, smoking and living in the North of the US is associated with higher insurance premiums. We assessed each model's predictive accuracy, evaluated limitations and highlighted potential improvements. We further outlined directions for extending our study.

## SECTION 2 - INTRODUCTION

With health care prices increasing rapidly and rising instances of new diseases, health insurance is considered to be a necessity for every household. Health insurance provides households with a financial safety net at times of medical emergencies. By buying health insurance policies, the insured agree to pay regular premiums and are guaranteed to be reimbursed for expenses incurred due to medical emergencies.

Actuaries price health insurance policies to help insurance companies stay afloat and make profits by collecting premiums exceeding the total amount paid to the insured. In order to do so, actuaries try to predict the expected medical bills which will be incurred in the future by insurance companies. To define the insurance premiums a number of factors are analysed including data on policy holders' socio-demographic characteristics. In theory, higher premiums are charged for those who are more likely to incur higher medical costs in the future.

In this study we aim to predict and explore factors impacting medical costs incurred by health insurance companies. For this we analyze cross-sectional social and demographic data from Kaggle, a public data platform. The data contains 1,338 observations which we used to build and analyze the five statistical models using multiple machine learning methods:

- **Model 1 - Baseline Model - *Multiple Linear Regression*** Model obtained using Best Subset model selection procedure. The Baseline Model will be used for comparison with the other four more sophisticated models below.
- **Model 2 - *Stochastic Gradient Descent*** Model.
- **Model 3 - *General Additive Model (GAM)*** using Smoothing Splines to achieve higher predictive accuracy while still retain a relatively high degree of model interpretability.
- **Model 4 - *High Dimensional*** Model built by adding interactions and applying transformations to the Baseline Model.
- **Model 5 - *Boosted Regression Trees*** Model to achieve higher predictive accuracy without emphasis on interpretability.

Seven characteristics **CHARGE**, **AGE**, **BMI**, **CHILDREN**, **SEX**, **SMOKER** and **REGION** were extracted from the Kaggle dataset. **CHARGE** was chosen to be the outcome variable. The rest of the variables were selected to be predictors. Please see more details on the variables in the tables below.

Table 1: OUTCOME VARIABLE

| Outcome variable | Description                                                        | Assumptions                                                                                                                                                                      |
|------------------|--------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CHA-RGE          | Individual medical costs billed by health insurance companies, USD | Medical costs billed by health insurance companies are correlated with the insurance premiums, the primary subject of our interest. USD are assumed to be the measurement units. |

Table 2: PREDICTOR VARIABLES

| # | Predictor | Description                                                                         | Type        | Assumptions                      |
|---|-----------|-------------------------------------------------------------------------------------|-------------|----------------------------------|
| 1 | AGE       | Age of policyholders, years                                                         | Continuous  |                                  |
| 2 | BMI       | Body mass index of policyholders, kg/m <sup>2</sup>                                 | Continuous  |                                  |
| 3 | CHILDREN  | Number of children/dependents covered by health insurance, 0 to 5                   | Categorical | Dependents include only children |
| 4 | SEX       | Gender of policyholders, male/female                                                | Categorical |                                  |
| 5 | SMOKER    | Smoking status, yes/no                                                              | Categorical |                                  |
| 6 | REGION    | Policyholders' residential area in the US, northeast, northwest/southeast/southwest | Categorical |                                  |

\*Please note that we treat CHILDREN as a continuous variable for Models 2, 3 and 5, as it takes six levels (integers from 0 to 5).

## SECTION 3 - EXPLORATORY DATA ANALYSIS

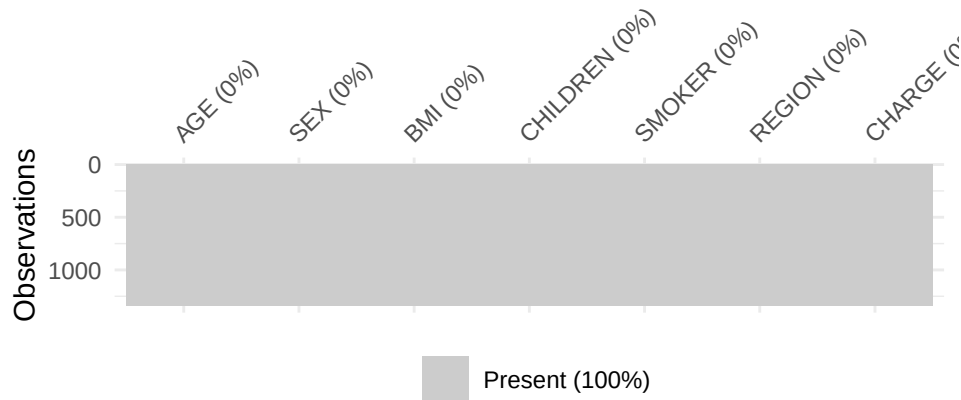
Exploratory data analysis (EDA) is carried out to determine the quality of the data that we will be working with:

- (i) identify any striking issues from general data overview and summary statistics,
- (ii) check for any missing data,
- (iii) examine presence of any interactions/need for any transformations based on scatterplots of the continuous predictors.

Step (i) - no striking issues, please see below.

```
##      AGE      SEX      BMI      CHILDREN
##  Min.   :18.00  Length:1338  Min.   :15.96  Min.   :0.000
## 1st Qu.:27.00  Class :character 1st Qu.:26.30 1st Qu.:0.000
## Median :39.00  Mode  :character  Median :30.40 Median :1.000
## Mean   :39.21                      Mean   :30.66 Mean   :1.095
## 3rd Qu.:51.00                      3rd Qu.:34.69 3rd Qu.:2.000
## Max.   :64.00                      Max.   :53.13 Max.   :5.000
##      SMOKER      REGION      CHARGE
## Length:1338      Length:1338      Min.   : 1122
## Class :character  Class :character 1st Qu.: 4740
## Mode  :character  Mode  :character Median : 9382
##                      Mean   :13270
##                      3rd Qu.:16640
##                      Max.   :63770
```

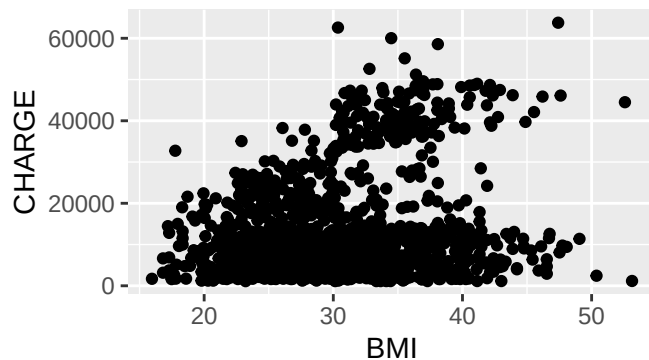
Step (ii) - no missing data, please see below.



Step (iii) - interactions present, transformations needed, please see below.

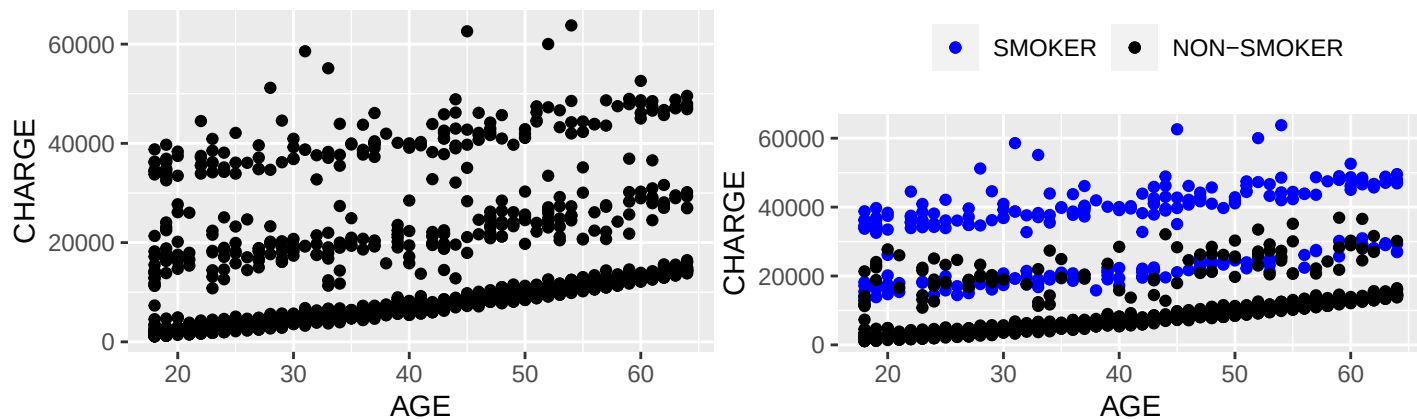
*Plot of CHARGE against BMI:*

We observe an approximate positive relationship, please see the below plot. However, the data points are scattered in a funnel shape suggesting that we may need to consider a transformation of BMI. This will be done in Model 4.



*Plot of CHARGE against AGE:*

We observe that the data lies in three approximately parallel lines, please refer to the plots below. This suggests that there may be other variables which lead to differences in medical costs for similar age groups. For example, plotting AGE against CHARGE while accounting for SMOKER illustrates the presence of the interaction. Therefore, we will explore further interaction terms in Model 4.



## SECTION 4 - DATA PREPARATION

The raw Kaggle data is now edited so that machine learning methods can be applied.

We obtain six datasets which will be used to train and test our models, see below. First, the original data is split into the training and test sets according to the 70/30 ratio: `Insurance_train` and `Insurance_test`. Then, for the purpose of building the five models and assessing their predictive accuracy, we transform each set into three datasets as follows: **original dataset**, **standardized dataset** and **recoded dataset**, please refer to the summary table below.

Table 3: DATASETS FOR BUILDING MODELS

| Model   | Training set used                                                  | Test set used                                                 |
|---------|--------------------------------------------------------------------|---------------------------------------------------------------|
| Model 1 | Recoded training dataset:<br><code>Insurance_train_recoded</code>  | Recoded test dataset:<br><code>Insurance_test_recoded</code>  |
| Model 2 | Standardized training dataset:<br><code>Insurance_train_std</code> | Standardized test dataset:<br><code>Insurance_test_std</code> |
| Model 3 | Original training dataset: <code>Insurance_train</code>            | Original test dataset: <code>Insurance_test</code>            |
| Model 4 | Recoded training dataset:<br><code>Insurance_train_recoded</code>  | Recoded test dataset:<br><code>Insurance_test_recoded</code>  |
| Model 5 | Original training dataset: <code>Insurance_train</code>            | Original test dataset: <code>Insurance_test</code>            |

Let us explain the steps we undertook to obtain the six datasets for the five models.

**Step 1 - load the libraries to perform data editing and split the data into a training/test set.**

```
library(recipes)
library(rsample)
set.seed(1) #obtain the 'same' 'random' training/test sets every time a code is run
train_index <- initial_split(Insurance_data, prop = 0.7)
train_data <- training(train_index) #training dataset
test_data <- testing(train_index) #test dataset
```

**Step 2 - build a recipe to convert categorical predictors into dummy variables and then obtain `Insurance_train` and `Insurance_test` datasets.**

Given that both the training and the test datasets contain the same number of variables (the sets differ only by the number of observations), we chose to create our initial recipe based on the training data.

```
initial_recipe <- recipe(CHARGE ~ ., data = train_data)
```

We keep in mind that the same recipe will later be applied to the test data as well.

Further, using the recipe, we convert three categorical variables `SMOKER`, `SEX` and `REGION` into dummy variables:

```
indicator_variables <- initial_recipe %>%
  step_dummy(all_nominal_predictors()) #all categorical variables converted into dummies
```

The following five new dummy variables were created:

Table 4: FIVE DUMMY VARIABLES

| Original Categorical Variable | Dummy Variable(s)                                    |
|-------------------------------|------------------------------------------------------|
| SMOKER, 2 levels              | SMOKER_yes                                           |
| SEX, 2 levels                 | SEX_male                                             |
| REGION, 4 levels              | REGION_northwest, REGION_southeast, REGION_southwest |

Update the recipe with the newly created dummy variables:

```
trained_recipe <- prep(indicator_variables, training = train_data)
```

Apply the recipe to the training/test datasets to obtain: `Insurance_train` and `Insurance_test` datasets which contain the five new dummy variables instead of the original SMOKER, SEX and REGION categorical variables.

```
Insurance_train <- bake(trained_recipe, new_data = train_data)
Insurance_test <- bake(trained_recipe, new_data = test_data)
```

**Step 3 - standardize predictors to create standardized training and test datasets:**  
`Insurance_train_std` and `Insurance_test_std`:

Given that all predictors are numeric, we scale and centre the predictors:

```
standardized_variables <- indicator_variables %>%
  step_center(all_numeric_predictors()) %>% #all numeric data have mean of 0
  step_scale(all_numeric_predictors()) #all numeric data have standard deviation of 1
```

Now we update the recipe with the standardized predictors:

```
further_trained_recipe <- prep(standardized_variables, training = train_data)
```

Apply the updated recipe to the training/test datasets to obtain the two standardized sets `Insurance_train_std` and `Insurance_test_std`:

```
Insurance_train_std <- bake(further_trained_recipe, new_data = train_data)
Insurance_test_std <- bake(further_trained_recipe, new_data = test_data)
```

**Step 4 - regroup CHILDREN and REGION into KIDS and REGION\_south, respectively, to obtain `Insurance_train_recoded` and `Insurance_test_recoded`.**

Regrouping will simplify our further analysis. Below see the summary of the new variables:

Table 5: REGROUPED BINARY VARIABLES

| Original Variable               | Re-grouped Binary Variable | Comments                                                                                |
|---------------------------------|----------------------------|-----------------------------------------------------------------------------------------|
| CHILDREN, categorical, 6 levels | KIDS                       | The binary variable KIDS takes 1 if CHILDREN $\geq 1$ and 0 otherwise.                  |
| REGION, categorical, 4 levels   | REGION_south               | he binary variable REGION_south takes 1 if REGION = southeast OR southwest, 0 otherwise |

CHILDREN into KIDS:

```
Insurance_train_recoded = (Insurance_train)
Insurance_train_recoded$KIDS[Insurance_train_recoded$CHILDREN == 0] <- 0
```

```
Insurance_train_recoded$KIDS[Insurance_train_recoded$CHILDREN >= 1] <- 1
Insurance_test_recoded = Insurance_test
Insurance_test_recoded$KIDS[Insurance_test_recoded$CHILDREN == 0] <- 0
Insurance_test_recoded$KIDS[Insurance_test_recoded$CHILDREN >= 1] <- 1
```

REGION into REGION\_south:

```
Insurance_train_recoded$REGION_south <- Insurance_train_recoded$REGION_southeast +
  Insurance_train_recoded$REGION_southwest
Insurance_test_recoded$REGION_south <- Insurance_test_recoded$REGION_southeast +
  Insurance_test_recoded$REGION_southwest
```

At this point we have performed all necessary steps for starting to create the models.

## SECTION 5 - FIVE MODELS

In this section we will build the five models of interest:

- Model 1: Multiple Linear Regression Model
- Model 2: Stochastic Gradient Descent Model
- Model 3: General Additive Model (GAM) using Smoothing Splines
- Model 4: High Dimensional Model
- Model 5: Boosted Regression Trees Model

### SUB-SECTION 5.1 - MODEL 1 - MULTIPLE LINEAR REGRESSION MODEL

To build a multiple linear regression model we choose to use the **Best Subset** variables selection method. **Best Subset** is used to ensure that all possible combinations of the six predictors are considered and we indeed obtain the best possible model. To perform Best Subset selection, we fit a least squares regression model for every possible combination. We then use adjusted  $R^2$ , Cp and BIC to select the best model.

Below we provide steps taken to obtain Model 1, interpret it and assess its predictive accuracy.

Table 6: Model 1 Summary

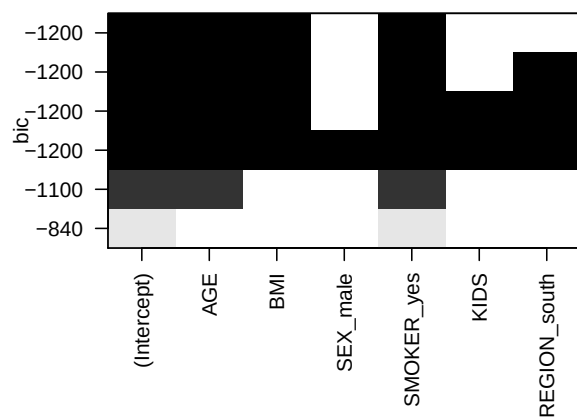
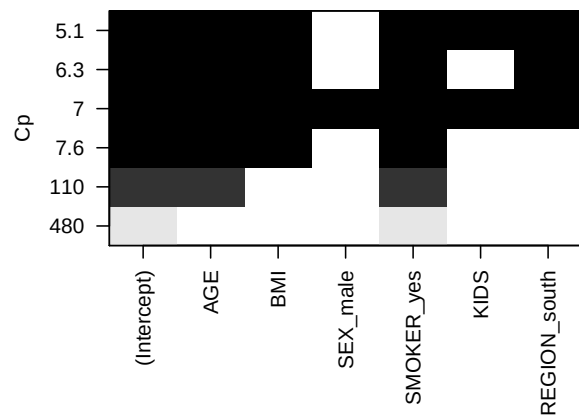
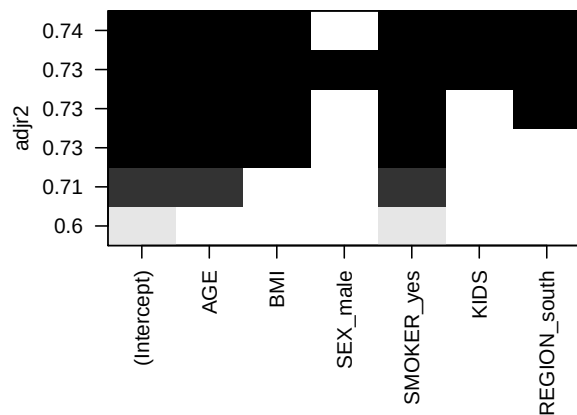
|                     |                                                                                      |
|---------------------|--------------------------------------------------------------------------------------|
| Model               | Model 1 - Multiple Linear Regression Model                                           |
| Variables Selection | Best Subset                                                                          |
| Method              |                                                                                      |
| Datasets Used       | Insurance_train_recoded, Insurance_test_recoded                                      |
| Comment             | KIDS and REGION_south are used instead of CHILDREN and REGION to ease interpretation |

**Step 1 - load the library which will be used to visualise the optimal variables based on the three criteria: Adjusted  $R^2$ , Cp and BIC**

```
library(leaps)
regfit.full <- regsubsets(CHARGE ~ . , data = Insurance_train_recoded, nvmax = 6)
reg.summary <- summary(regfit.full)
```

**Step 2 - perform best subset selection using the Adjusted  $R^2$ , Cp and BIC criteria to obtain the best model.**

The best subsets method, under all three criteria suggests that a model excluding the variable SEX may be considered as the best model:



### Step 3 - Model 1 analysis and interpretation.

From the below summary statistics, the p values show that the continuous variables AGE and BMI as well as the binary variable SMOKER\_yes are highly significant with p values less than  $2 \times 10^{-16}$ . The binary variables KIDS and REGION\_south are significant at the 10% level. The F statistic was found to be 520 with a p-value of less than  $2 \times 10^{-16}$ .

These allows us to conclude that, overall, Model 1 has a high explanatory power:

```
linear_regression_model <- lm(CHARGE ~ AGE + BMI + SMOKER_yes + KIDS + REGION_south,
                             data = Insurance_train_recoded)
summary(linear_regression_model)
```

```
##
## Call:
## lm(formula = CHARGE ~ AGE + BMI + SMOKER_yes + KIDS + REGION_south,
##     data = Insurance_train_recoded)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10661  -3133  -1077    1555   30425
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
```



```
## (Intercept)  -12697.38    1166.67 -10.883    <2e-16 ***
## AGE          260.20      14.67  17.736    <2e-16 ***
## BMI          355.86      34.98  10.173    <2e-16 ***
## SMOKER_yes   23105.34    495.62  46.619    <2e-16 ***
## KIDS         730.91     412.48   1.772    0.0767 .
## REGION_south -752.51     418.38  -1.799    0.0724 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6234 on 930 degrees of freedom
## Multiple R-squared:  0.7365, Adjusted R-squared:  0.7351
## F-statistic: 520 on 5 and 930 DF, p-value: < 2.2e-16
```

Based on the regression coefficients, we observe holding all other variables fixed, that:

- **AGE**: Medical costs billed by health insurance tend to increase with age. Indeed as an individual gets older, they are closer to their life expectancy and therefore insurance premiums tend to increase<sup>1</sup>.
- **BMI**: Similarly, medical costs billed by health insurance tend to increase with BMI. Having a higher BMI can be an indicator for obesity which puts individuals at a higher risk for certain serious illnesses such as type II diabetes, heart disease etc. And therefore it is likely for an individual with a high BMI to have higher medical costs<sup>2</sup>.

Comparing the coefficients for the two continuous variables, it can be observed that the coefficient for BMI is larger than the coefficient for AGE. Therefore, it can be inferred that BMI has a higher association with medical costs billed by health insurance.

- **SMOKER\_yes**: The coefficient for SMOKER\_yes is positive indicating that an individual who is a smoker has, on average, higher medical costs. Indeed smoking has proven to be a major cause of serious illnesses such as cancer, heart disease, etc<sup>3</sup>.

Note that SMOKER\_yes' coefficient has the highest absolute value, indicating that whether an individual smokes or not has the highest association with medical costs billed by health insurance compared to the rest of the binary variables.

- **KIDS**: Similarly, individuals who have at least one dependent tend to have higher medical costs. This is expected as the dependents are also covered by health insurance and therefore the medical costs for multiple individuals are taken into account.
- **REGION\_south**: The coefficient for REGION\_south is negative, indicating that an individual based in the North of the US has, on average, higher medical costs.

## Step 5 - assessing prediction accuracy of Model 1.

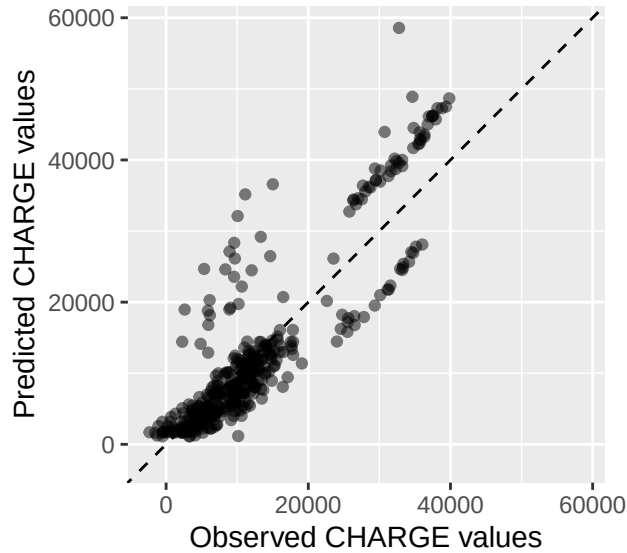
**Root Mean Squared Error**(RMSE) will be used to assess prediction accuracy throughout the report. The RMSE is the standard deviation of the residuals (i.e. prediction errors), measuring how far the true observations are from the predicted ones. In other words, the RMSE shows how spread out the residuals are. The higher RMSE is, the more likely it is that insurance premiums will be over or under-estimated. Based on the test dataset we calculate the RMSE for Model 1 and find it to be USD 5,688.

Below is a plot illustrating predicted **CHARGE** values against the observed ones. The closer the points lie to the 45° line, the more accurate the predictions are.

<sup>1</sup>Estevez, E. 2021, *How Age Affects Life Insurance*, viewed 2 February, 2022, <https://www.investopedia.com/articles/personal-finance/022615/how-age-affects-life-insurance-rates.asp>

<sup>2</sup>LifeSEARCH, 2022. *What is BMI and why is it important?*, viewed February 2, 2022, <https://www.lifesearch.com/advice-and-guides/what-is-bmi-and-why-is-it-important>

<sup>3</sup>Izumi, Y., Tsuji, I., Ohkubo, T., Kuwahara, A., Nishino, Y., Hisamichi, S., 2001. *Impact of smoking habit on medical care use and its costs: a prospective observation of National Health Insurance beneficiaries in Japan*, viewed February 2, 2022, <https://academic.oup.com/ije/article/30/3/616/736943>



The RMSE value of USD 5,688 suggests that the predictive accuracy of the model is not as good. Further examining the plot above, it can be observed that the predicted values lie closely to the observed values for small values of CHARGE (below USD 20,000) suggesting that the predictive accuracy for CHARGE is better for smaller values than it is for larger values.

---

**CONCLUSION:** we arrived at the Multiple Linear Regression Model 1:

$$\text{CHARGE} = -12,697 + 260.2(\text{AGE}) + 355.86(\text{BMI}) - 752.51(\text{REGION\_south}) + 730.91(\text{KIDS}) + 23,105.34(\text{SMOKER\_yes})$$

By analyzing the coefficients of the model, we conclude that being older, having a higher BMI, living in the North of the US, having children/dependents and smoking were are five characteristics associated with higher medical costs billed by health insurance providers.

By looking at the RMSE value and the plot of the predicted CHARGE values versus the observed ones, we infer that the predictive accuracy is not the best and should be improved, especially for larger values of CHARGE. These will be further examined in Models 2 to 5.

Model 1 will be used as a baseline for making comparisons with the Models 2 to 5.

## SUB-SECTION 5.2 - MODEL 2 - STOCHASTIC GRADIENT DESCENT MODEL

Our Model 2 is a multiple linear regression model built using Stochastic Gradient Descent (SGD). We chose to use SGD rather than Gradient Descent because we consider the Insurance dataset to have a relatively large number of observations, which will allow for our model to learn faster in comparison to the Gradient Descent approach<sup>4</sup>.

To perform stochastic gradient descent calculations, we created separate dataframes for the predictor variables and the outcome variable. The standardized datasets: `Insurance_train_std` and `Insurance_test_std` are used. Using standardized data allows us to avoid biasing our Model 2. Such bias is possible to occur as in the SGD calculation we will constantly update values of the model parameters (as per different random samples)<sup>5</sup>.

<sup>4</sup>Dutta, S., 2019. *Why Stochastic Gradient Descent works?*, viewed 2 February, 2022, <https://towardsdatascience.com/https-towardsdatascience-com-why-stochastic-gradient-descent-works-9af5b9de09b8>

<sup>5</sup>Loukas, S., 2020, *How and why to Standardize your data: A python tutorial*, viewed February 2, 2022, <https://towardsdatascience.com/how-and-why-to-standardize-your-data-996926c2c832>

```

pred_train <- Insurance_train_std[, -4] #training data for all predictors
out_train <- Insurance_train_std[, 4] #training data for outcome CHARGE
predictor_test <- Insurance_test_std[, -4] #test data for all predictors
outcome_test <- Insurance_test_std[, 4] #test data for outcome CHARGE

```

**Step 1 - add a column of 1s' to the training data for predictors to account for an intercept term and specify the loss function.**

```

int <- rep(1, nrow(pred_train)) #creating a column of 1's for the intercept term
pred_int_train <- cbind(int, pred_train) #adding the above column to predictor training data

```

To execute SGD, we first specify our loss function. The loss function that we will use is the altered Mean-Squared Error function. This is used to ensure the gradient of our chosen loss function is not a constant. Please find the expression for the function below.

$$\frac{1}{2n}(Y - X\beta^T)^T(Y - X\beta^T)$$

Where:

- $n$  is the number of observations
- $Y$  is the vector of observed values of CHARGE
- $X\beta^T$  is vector of predicted values of CHARGE
- $\beta$  is the vector of the regression parameters

Next, we need to minimize the above loss function. We do so by calculating its partial derivative with respect to each parameter in the regression model. We obtain the following expression for the partial derivative with respect to  $\beta$ :

$$\frac{1}{n}X^T(X\beta^T - Y)$$

**Step 2 - define a function to calculate the partial derivative of the loss function.**

In this step we will write a code<sup>6</sup> to obtain the above partial derivative:

```

loss_par_deriv <- function(pred_int_train, out_train, beta) {
  part_deriv <- (1/nrow(out_train))*t(pred_int_train) %*% ((pred_int_train %*% t(beta))
                                                            - out_train))
  return(t(part_deriv))
}

```

**Step 3 - define the SGD function.**

Next we will construct the SGD function in which we randomly produce various smaller samples each containing 5 observations. We then iterate over these smaller samples 10,000 times. Note that  $\beta$  is updated after each iteration.

The SGD function<sup>7</sup> is as follows. It will be used in Step 4 for further SGD calculation.

```

stoch_gradDescent <- function(pred_train, out_train, alpha, n){
  #combining predictor matrix and outcome vector to draw random samples from:
  all_train <- data.frame(cbind(out_train, pred_int_train))
  #defining beta to store values of regression coefficients:

```

<sup>6</sup>Topor, J., 2016 *Multivariate Linear Regression Analysis using Stochastic Gradient Descent*, viewed February 2, 2022, [https://rpubs.com/jt\\_rpubs/279269](https://rpubs.com/jt_rpubs/279269)

<sup>7</sup>Topor, J., 2016 *Multivariate Linear Regression Analysis using Stochastic Gradient Descent*, viewed February 2, 2022, [https://rpubs.com/jt\\_rpubs/279269](https://rpubs.com/jt_rpubs/279269)

```

beta <- matrix(c(0, 0, 0, 0, 0, 0, 0, 0, 0), nrow = 1)
#defining a matrix containing updated beta values after each iteration:
beta_upd <- matrix(NA, nrow = n, ncol = 9)
set.seed(1)

#iterate over smaller random samples and update value of beta after each iteration:
for (i in 1:n) {
  #sampling 5 observations with replacement from our data frame:
  all_train_sample <- as.matrix( all_train[sample(nrow(all_train), 5, replace = TRUE), ])
  #obtaining predictor variable data from random samples:
  pred_sample <- as.matrix(all_train_sample[,2:10])
  #obtaining outcome variable data from random samples:
  out_sample <- as.matrix(all_train_sample[,1])
  #updating beta using random samples:
  beta <- beta - alpha * loss_par_deriv(pred_sample, out_sample, beta)
  #saving the theta values for each iteration:
  beta_upd[i,] <- beta
}
return(beta_upd)
}

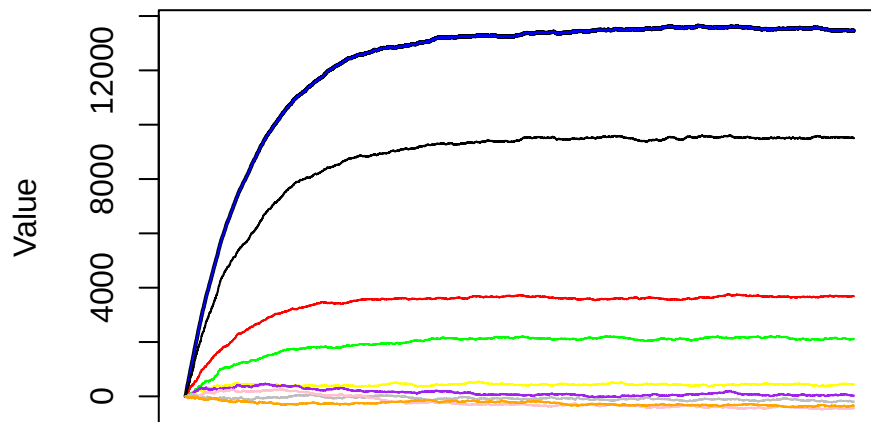
```

#### Step 4 - SGD calculation

First please note that the learning rate is denoted by  $\alpha$  in the SGD function.  $\alpha$  is a hyper-parameter that determines how quickly we descent down the gradient of the loss function<sup>8</sup>. In this model we set the learning rate  $\alpha$  to be 0.001 and implement SGD:

```
result <- stoch_gradDescent(pred_train, out_train, 0.001, 10000)
```

Then, we plot the above result to verify if convergence of the **beta** parameters can be achieved within 10,000 iterations.



$\alpha = 1$

From the plot above we see that indeed the convergence is achieved. Hence, the Model 2 will now be of the

<sup>8</sup>Zulkifli, H., 2018, *Understanding Learning Rates and How It Improves Performance in Deep Learning*, viewed February 2, 2022, <https://towardsdatascience.com/understanding-learning-rates-and-how-it-improves-performance-in-deep-learning-d0d4059c1c10>

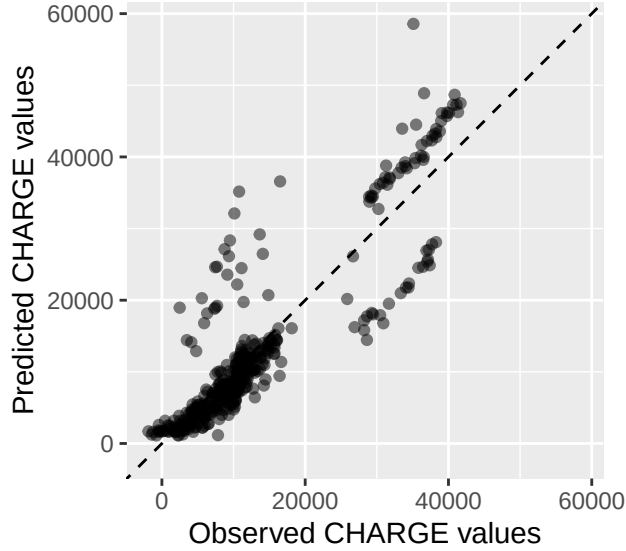
following form:

$$\begin{aligned} \text{CHARGE} = & 13463.62 + 3687.88(\text{AGE}) + 2114.52(\text{BMI}) + 445.64(\text{CHILDREN}) + 30.69(\text{SEX\_male}) \\ & + 9513.44(\text{SMOKER\_yes}) - 191.61(\text{REGION\_northwest}) - 432.17(\text{REGION\_southeast}) - 346.78(\text{RE-} \\ & \text{GION\_southwest}) \end{aligned}$$

#### Step 5 - assessing prediction accuracy.

Based on the standardized test dataset we calculate the RMSE for Model 2 and find it to be USD 5,514.

Then we plot predicted **CHARGE** values against the observed ones for further illustration:



**CONCLUSION:** the following linear regression Model 2 using SGD is as follows:

$$\begin{aligned} \text{CHARGE} = & 13463.62 + 3687.88(\text{AGE}) + 2114.52(\text{BMI}) + 445.64(\text{CHILDREN}) + 30.69(\text{SEX\_male}) \\ & + 9513.44(\text{SMOKER\_yes}) - 191.61(\text{REGION\_northwest}) - 432.17(\text{REGION\_southeast}) - 346.78(\text{RE-} \\ & \text{GION\_southwest}) \end{aligned}$$

Using SGD allowed us to achieve an approximately 3% improvement in predictive performance compared to Model 1.

### SUB-SECTION 5.3 - MODEL 3 - GENERALIZED ADDITIVE MODEL (GAM)

In this section a flexible linear approach that preserves interpretability will be used to build the Model 3. Building up on the multiple linear regression method implemented in Model 1, we apply the more general GAM framework, which allows for non-linear functions of each variable to be considered. This allows us to avoid manual variables transformation and ensure that no non-linear relationships are missed. Below you will find the steps we undertook to build, analyze and interpret Model 3.

The general GAM form of Model 3 is as follows:

$$\text{CHARGE} = \text{beta\_0} + f\_1(\text{BMI}) + f\_2(\text{AGE}) + f\_3(\text{REGION\_northwest}) + f\_4(\text{REGION\_southeast}) + f\_5(\text{REGION\_southwest}) + f\_6(\text{CHILDREN}) + f\_7(\text{SMOKER\_yes}) + \text{epsilon}$$

Further, **Insurance\_train** and **Insurance\_test** datasets with the ungrouped **CHILDREN** and **REGION** predictors are used to build Model 3. This is because we choose to forego some degree of interpretability and fully utilize the information provided by **CHILDREN** and **REGION** to improve the model's predictive accuracy.

**Step 1 - load the library containing GAM tools and obtain the GAM baseline model.**

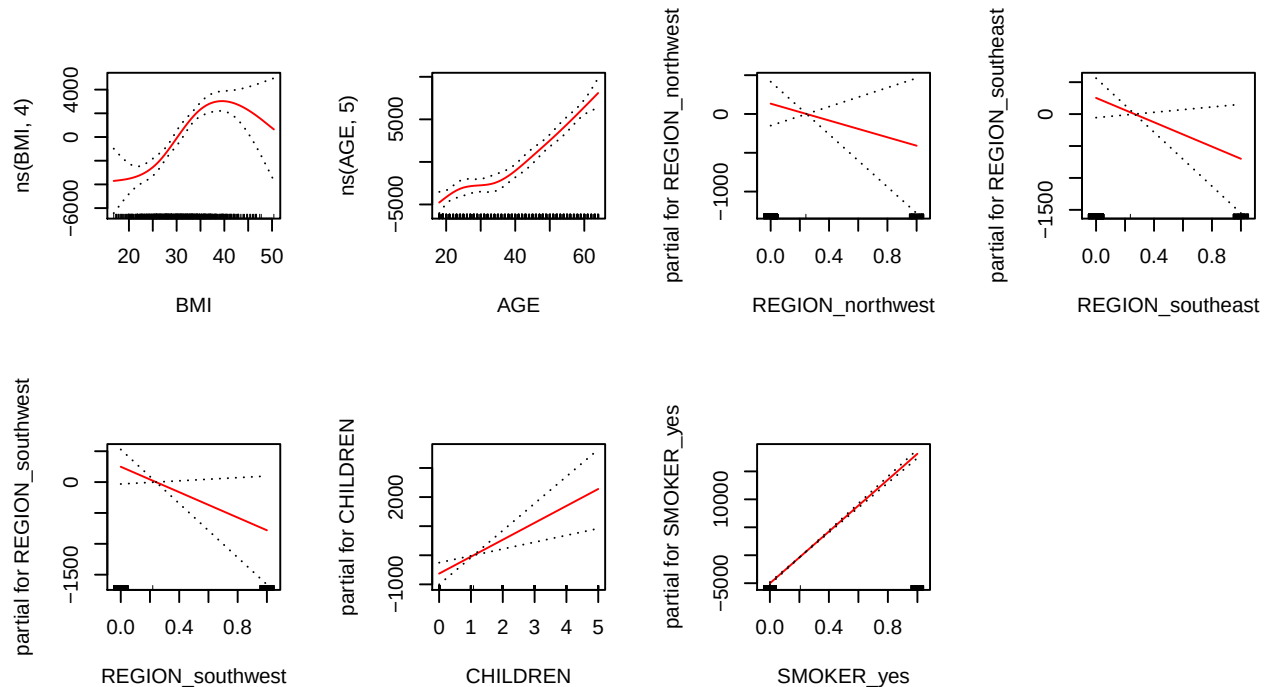
```
library(gam)
```

Below is the GAM baseline model using natural splines.

```
gam_baseline = lm(CHARGE ~ ns(BMI, 4) + ns(AGE, 5) + REGION_northwest + REGION_southeast
                  + REGION_southwest + CHILDREN + SMOKER_yes, data = Insurance_train)
```

The model is essentially a slight extension of our linear regression Model 1 with the only difference of using appropriate basis functions for the natural splines. Please note the variable **SEX** was excluded as per the analysis in step 2, SUB-SECTION 5.1.

To determine if **AGE** and **BMI** could be assumed linear we resort to the below plots. From them we see that while **AGE** may be assumed to be linear, **BMI** is highly non-linear.



**Step 2 - build and compare GAM candidate models.**

Based on the nature of the variables **AGE** and **BMI**, let us construct the GAM candidate models:

```
#Candidate Model 1 - AGE and BMI non-linear:
gam_model_1 = gam(CHARGE ~ s(BMI, 4) + s(AGE, 5) + REGION_northwest + REGION_southeast +
                  REGION_southwest + CHILDREN + SMOKER_yes, data = Insurance_train)
#Candidate Model 2 - BMI non-linear:
gam_model_2 = gam(CHARGE ~ s(BMI, 4) + AGE + REGION_northwest + REGION_southeast +
                  REGION_southwest + CHILDREN + SMOKER_yes, data = Insurance_train)
#Candidate Model 3 - AGE non-linear:
gam_model_3 = gam(CHARGE ~ BMI + s(AGE,5) + REGION_northwest + REGION_southeast +
                  REGION_southwest + CHILDREN + SMOKER_yes, data = Insurance_train)
```

Comparing the GAM candidate models using the ANOVA approach:

```
anova(gam_baseline, gam_model_1, gam_model_2, gam_model_3, test = "F")
```

```
## Analysis of Variance Table
##
```

```
## Model 1: CHARGE ~ ns(BMI, 4) + ns(AGE, 5) + REGION_northwest + REGION_southeast +
## REGION_southwest + CHILDREN + SMOKER_yes
## Model 2: CHARGE ~ s(BMI, 4) + s(AGE, 5) + REGION_northwest + REGION_southeast +
## REGION_southwest + CHILDREN + SMOKER_yes
## Model 3: CHARGE ~ s(BMI, 4) + AGE + REGION_northwest + REGION_southeast +
## REGION_southwest + CHILDREN + SMOKER_yes
## Model 4: CHARGE ~ BMI + s(AGE, 5) + REGION_northwest + REGION_southeast +
## REGION_southwest + CHILDREN + SMOKER_yes
## Res.Df      RSS      Df Sum of Sq      F      Pr(>F)
## 1      921 3.4893e+10
## 2      921 3.4726e+10  0.00024809 166830397 17834.7597 4.585e-06 ***
## 3      925 3.5380e+10 -4.00012536 -654022444      4.3364  0.001765 **
## 4      924 3.5352e+10  1.00000263  28070482      0.7445  0.388452
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Based on the ANOVA output above, there is significant evidence that the Candidate Model 1 with non-linear functions of AGE and BMI is superior to the rest of the candidate models (judged by p-values). Hence, the Candidate Model 1 is chosen to be our Model 3, the GAM model using smoothing splines.

### Step 3 - Model 3 analysis and interpretation.

Let's perform the analysis of Model 3.

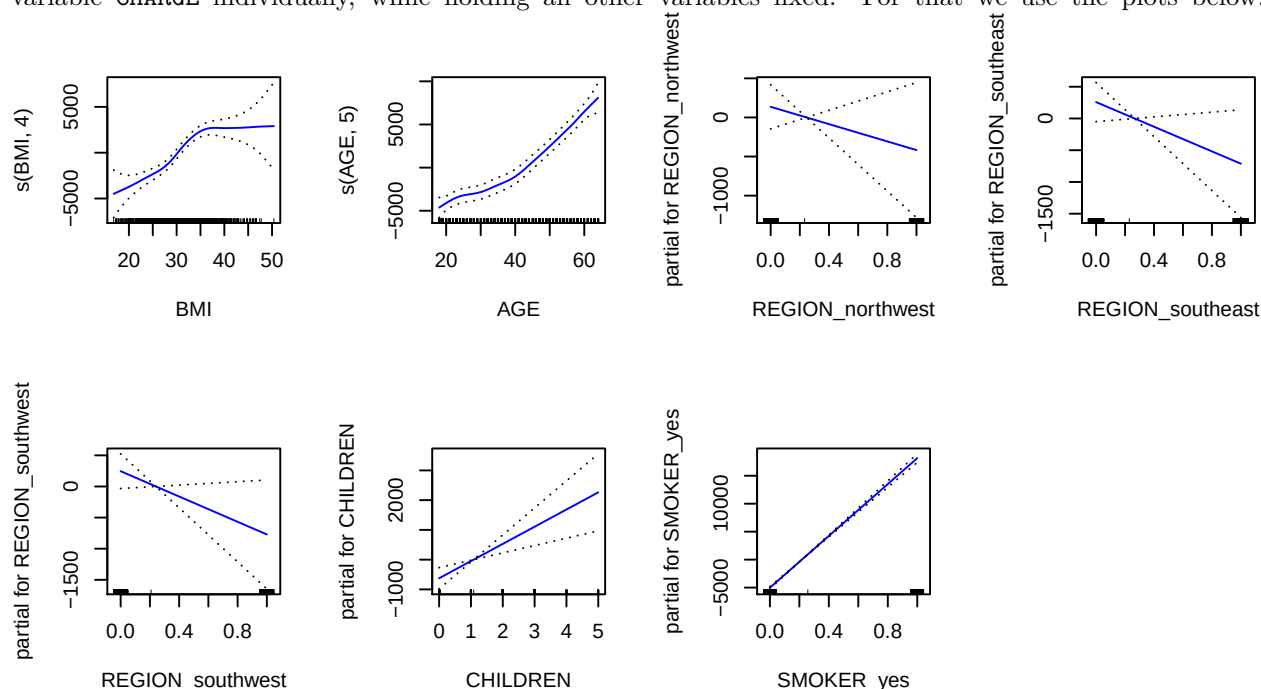
```
summary(gam_model_1)
```

```
##
## Call: gam(formula = CHARGE ~ s(BMI, 4) + s(AGE, 5) + REGION_northwest +
## REGION_southeast + REGION_southwest + CHILDREN + SMOKER_yes,
## data = Insurance_train)
## Deviance Residuals:
##      Min       1Q   Median       3Q      Max
## -10929.4  -3767.8   -913.7   1542.8  31228.5
##
## (Dispersion Parameter for gaussian family taken to be 37704369)
##
##      Null Deviance: 137192579302 on 935 degrees of freedom
## Residual Deviance: 34725714947 on 920.9998 degrees of freedom
## AIC: 19001.92
##
## Number of Local Scoring Iterations: NA
##
## Anova for Parametric Effects
##              Df      Sum Sq   Mean Sq    F value    Pr(>F)
## s(BMI, 4)      1 4.3455e+09 4.3455e+09  115.2525 < 2.2e-16 ***
## s(AGE, 5)      1 1.1499e+10 1.1499e+10  304.9834 < 2.2e-16 ***
## REGION_northwest 1 8.6874e+07 8.6874e+07    2.3041  0.12938
## REGION_southeast 1 1.8147e+08 1.8147e+08    4.8129  0.02850 *
## REGION_southwest 1 1.5945e+08 1.5945e+08    4.2290  0.04002 *
## CHILDREN      1 9.0064e+08 9.0064e+08   23.8870 1.205e-06 ***
## SMOKER_yes     1 8.4694e+10 8.4694e+10 2246.2724 < 2.2e-16 ***
## Residuals     921 3.4726e+10 3.7704e+07
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Anova for Nonparametric Effects
```

```
##              Npar Df Npar F      Pr(F)
## (Intercept)
## s(BMI, 4)      3 5.5416 0.0009014 ***
## s(AGE, 5)      4 4.4218 0.0015184 **
## REGION_northwest
## REGION_southeast
## REGION_southwest
## CHILDREN
## SMOKER_yes
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The results of ANOVA for Parametric effects from the summary table above suggest that choosing model with the non-linear AGE and BMI was the right decision to make, as the p-values for AGE and BMI are extremely low (both  $<0.0015$ ).

Given the additive nature of GAMs, we can easily examine the effects of each predictor on the response variable CHARGE individually, while holding all other variables fixed. For that we use the plots below.



Based on the analysis of the plots, we infer the following:

- **AGE**: holding all other variables fixed, medical costs billed by health insurance tend to increase with age. This relationship was also observed in Model 1.
- **BMI**: in general, medical costs billed by health insurance rise as one's BMI increases. Whilst this trend was also observed in Model 1, more details are revealed by the GAM Model 3. In particular, based on the visual inspection of the graph, the rate at which the costs are rising is higher for BMI values of roughly less than 35. The explanation for this could be the following. According to the Centres for Disease Control and Prevention, people with BMI above 35 could be classified as having class two obesity<sup>9</sup>. It could be argued that past a certain point obesity-related health issues become very similar, meaning that a person becoming even more obese will lead to a less than proportional increase in his or her medical bills.

<sup>9</sup>CDC, 2021, *Defining Adult Overweight and Obesity*, viewed 2 February 2022, <https://www.cdc.gov/obesity/adult/defining.html>

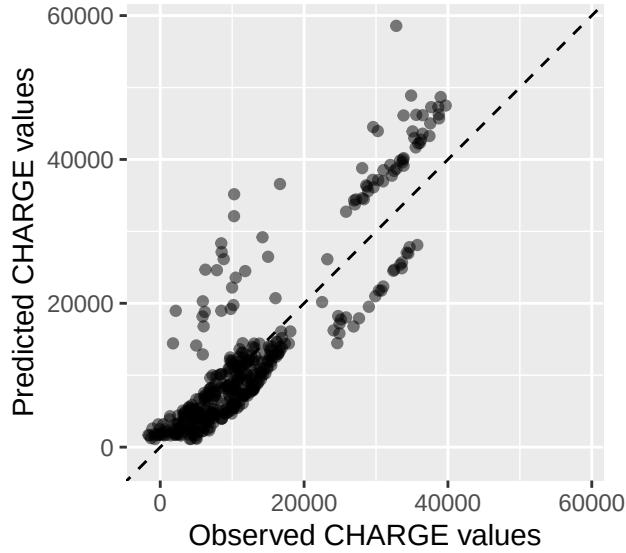


- ***SMOKER\_yes, CHILDREN***: similarly to what was found in Model 1, being a smoker and having more dependents tend to increase medical costs incurred by health insurance.
- ***REGION\_southeast, REGION\_southwest, REGION\_northwest***: living in the northeast of the US is associated with higher medical costs compared to all the other US regions.

#### Step 4 - Assessing prediction accuracy.

Based on the test set we calculate the RMSE for Model 3 and find it to be USD 5,649.

Then we plot predicted **CHARGE** values against the observed ones for further illustration:



As seen from the plot, the data points are more closely scattered around the 45° line, indicating an improvement in the model's predictive accuracy compared to Models 1 and 2 while retaining some degree of interpretability.

---

**CONCLUSION:** we obtained a relatively interpretable and flexible GAM model 3. Yet, the drawback of this approach is that important interactions between predictors may be missed, such as, for instance, an interaction between **AGE** and **SMOKER\_yes**. We attempt to address this issue in Model 4. Furthermore, an even more flexible approach using regression trees will be considered in Model 5.

### SUB-SECTION 5.4 - MODEL 4 - HIGH DIMENSIONAL MODEL

A high-dimensional model is sought with neither particular predictive accuracy nor interpretability goals in mind. The model will be built based on the the baseline Model 1 obtained in SUB-SECTION 5.1:

$$CHARGE = -12697 + 260.2(AGE) + 355.86(BMI) - 752.51(REGION\_south) + 730.91(KIDS) + 23,105.34(SMOKER\_yes)$$

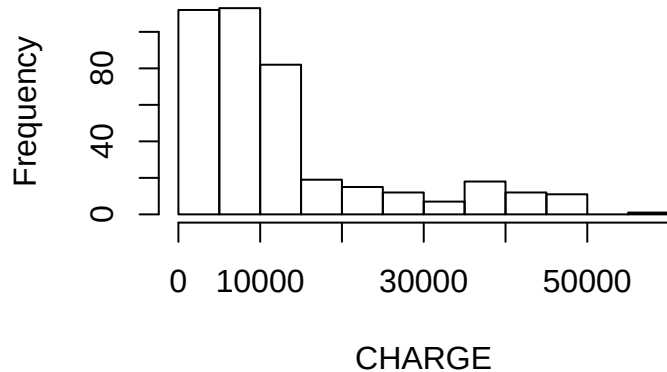
As only five predictor variables are used in the baseline model, we choose to increase the dimensionality of the model by considering non-linear transformations of predictor variables and the outcome variable as well as different interaction terms between the predictors. Suitable transformations and interaction terms are assessed on the basis of Adjusted  $R^2$ . Increase in Adjusted  $R^2$  indicates an improvement in a model while accounting for the number of terms included in the model.

Let us build Model 4.

### Step 1 - transforming the outcome variable CHARGE.

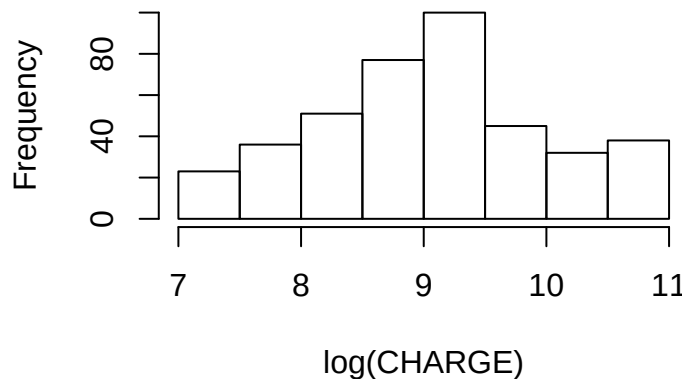
First, let us plot a histogram of the outcome variable CHARGE to see if there are any issues.

#### Histogram of CHARGE



It is clear from the histogram above that the outcome variable CHARGE is highly skewed, so a log transformation is applied to make the distribution of CHARGE more normal. Please see the plot below.

#### Histogram of log(CHARGE)



Let us now consider a Candidate 1 Model with a transformed outcome variable:

```
Candidate_1 = lm(log(CHARGE) ~ AGE + BMI + KIDS + SMOKER_yes + REGION_south,  
                 data = Insurance_train_recoded)
```

Compared to the baseline Model 1, using log(CHARGE) instead of CHARGE leads to an improvement of the model as adjusted  $R^2$  rises from 74.93% to 75.19%.

### Step 2 - transforming continuous predictor variables AGE and BMI.

Now, we transform each of the two continuous predictor variables in the model. We seek to choose the best transformation on each X predictor variable among this list: log(X), exp(X), sqrt(X),  $X^2$ ,  $X^3$ ,  $X^4$ ,  $X^5$ .

We explore the correlations between log(CHARGE) and each transformed predictor variable. The transformation which yields the largest correlation with log(CHARGE) is then chosen because this indicates the strongest linear relationship between log(CHARGE) and the transformed variable.

The largest correlations obtained with log(CHARGE) are with log(AGE) and log(BMI):

```
cor(log(CHARGE), log(AGE))
```

```
## [1] 0.536225
```

```
cor(log(CHARGE), log(BMI))
```

```
## [1] 0.1905207
```

Upon adding the transformations  $\log(\text{AGE})$ ,  $\log(\text{BMI})$  separately as well as simultaneously Candidate Model 2 with only  $\log(\text{BMI})$  transformation results in the largest improvement in adjusted  $R^2$  from 75.19% to 75.24%:

```
Candidate_2 = lm(log(CHARGE) ~ AGE + log(BMI) + KIDS + SMOKER_yes + REGION_south,
  data = Insurance_train_recoded)
```

### Step 3 - interaction terms.

One by one, we assess the changes in Adjusted  $R^2$  when adding interaction terms to Candidate Model 2. Please see the table below, which consists of the most promising significant interaction terms:

Table 7: MOST PROMISING SIGNIFICANT INTERACTION TERMS

| Interaction                    | Adjusted $R^2$ |
|--------------------------------|----------------|
| AGE*SMOKER_yes                 | 80.27%         |
| $\log(\text{BMI})$ *SMOKER_yes | 77.66%         |
| AGE*KIDS                       | 76.66%         |
| KIDS*SMOKER_yes                | 76.83%         |
| SMOKER_yes*REGION_south        | 76.60%         |
| AGE*REGION_south               | 76.35%         |

### Step 4 - obtaining a high-dimensional Model 4.

Let us add all significant interaction terms to increase dimensionality of model:

```
Candidate_3 = lm(log(CHARGE) ~ AGE + log(BMI) + KIDS + SMOKER_yes + REGION_south
  + AGE*KIDS + AGE*SMOKER_yes + AGE*REGION_south + log(BMI)*SMOKER_yes +
  KIDS*SMOKER_yes + SMOKER_yes*REGION_south,
  data = Insurance_train_recoded )
```

The issue here is that the p-value of  $\log(\text{BMI})$  is substantially larger than 0.05, indicating that it is a highly insignificant variable in the model. Let us therefore remove  $\log(\text{BMI})$  along with the interaction term that includes it ( $\log(\text{BMI})$ \*SMOKER\_yes) to obtain Candidate Model 4.

```
Candidate_4 = lm(log(CHARGE) ~ AGE + KIDS + SMOKER_yes + REGION_south + AGE*KIDS
  + AGE*SMOKER_yes + AGE*REGION_south + KIDS*SMOKER_yes
  + SMOKER_yes*REGION_south,
  data = Insurance_train_recoded) #removed log(BMI) and log(BMI)*SMOKER_yes
summary(Candidate_4)
```

```
##
```

```
## Call:
```

```
## lm(formula = log(CHARGE) ~ AGE + KIDS + SMOKER_yes + REGION_south +
```

```
## AGE * KIDS + AGE * SMOKER_yes + AGE * REGION_south + KIDS *
```

```
## SMOKER_yes + SMOKER_yes * REGION_south, data = Insurance_train_recoded)
```

```
##
```

```
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.67054 -0.19149 -0.09574  0.02334  2.45726
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    7.077405   0.070830  99.921 < 2e-16 ***
## AGE            0.041249   0.001673  24.658 < 2e-16 ***
## KIDS           0.645542   0.084071   7.679 4.08e-14 ***
## SMOKER_yes     2.645214   0.109111  24.243 < 2e-16 ***
## REGION_south  -0.389400   0.082349  -4.729 2.61e-06 ***
## AGE:KIDS       -0.008732   0.001982  -4.406 1.18e-05 ***
## AGE:SMOKER_yes -0.028284   0.002433 -11.625 < 2e-16 ***
## AGE:REGION_south 0.006203   0.001941   3.195 0.00144 **
## KIDS:SMOKER_yes -0.323292   0.067898  -4.761 2.23e-06 ***
## SMOKER_yes:REGION_south 0.278025   0.066328   4.192 3.03e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4162 on 926 degrees of freedom
## Multiple R-squared:  0.7981, Adjusted R-squared:  0.7961
## F-statistic: 406.6 on 9 and 926 DF,  p-value: < 2.2e-16
```

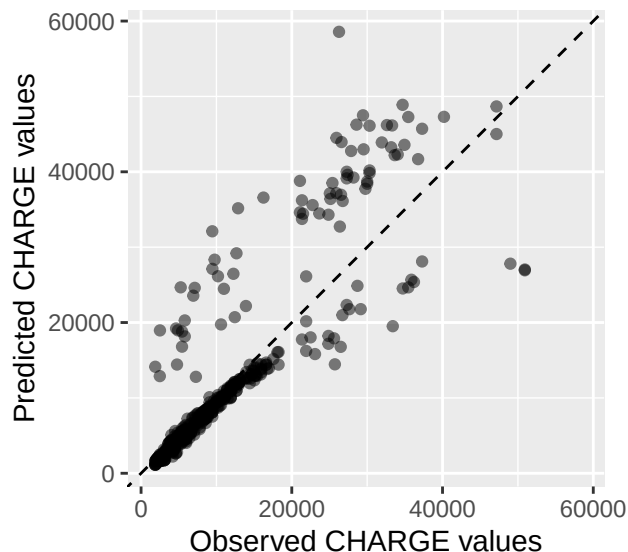
All variables in Candidate Model 4 are indeed now significant, and we have maximized the number of interaction terms which can be included in our target Model 4. Therefore, this model is the best high-dimensional model we could have obtained and we choose it to be our final High-dimensional Model 4 with 10 terms:

$$\log(\text{CHARGE}) = 7.08 + 0.04(\text{AGE}) + 0.65(\text{KIDS}) + 2.65(\text{SMOKER\_yes}) - 0.39(\text{REGION\_south}) - 0.009(\text{AGE} \times \text{KIDS}) - 0.03(\text{AGE} \times \text{SMOKER\_yes}) + 0.006(\text{AGE} \times \text{REGION\_south}) - 0.32(\text{KIDS} \times \text{SMOKER\_yes}) + 0.28(\text{SMOKER\_yes} \times \text{REGION\_south})$$

### Step 5 - assessing prediction accuracy.

Based on the test set we calculate the RMSE for Model 1 and find it to be USD 6,450.

Then we plot predicted CHARGE values against the observed ones for further illustration:



---

**CONCLUSION:** The high dimensional Model 4 is obtained:  $\log(\text{CHARGE}) = 7.08 + 0.04(\text{AGE}) + 0.65(\text{KIDS}) + 2.65(\text{SMOKER\_yes}) - 0.39(\text{REGION\_south}) - 0.009(\text{AGE}*\text{KIDS}) - 0.03(\text{AGE}*\text{SMOKER\_yes}) + 0.006(\text{AGE}*\text{REGION\_south}) - 0.32(\text{KIDS}*\text{SMOKER\_yes}) + 0.28(\text{SMOKER\_yes}*\text{REGION\_south})$

While we achieved our goal of increasing Model 1's dimensionality by introducing interaction terms and transformations, we nevertheless note there is a risk of multicollinearity, which should be assessed further, please refer to SECTION 6, LIMITATIONS AND IMPROVEMENTS.

The model is relatively suitable for interpretational purposes as the relationships between the predictors and the outcome variable **CHARGE** are similar to those discovered in Model 1. At the same time, we recognize that Model 4's predictive accuracy is relatively low and could be potentially improved.

## SUB-SECTION 5.5 - MODEL 5 - BOOSTED REGRESSION TREES MODEL

Here boosting is applied on the basis of regression trees to achieve a model with higher predictive accuracy at the expense of interpretability. To account for complex relationships between the predictors and the response variable, multiple trees are produced and then combined yielding a single prediction. Below we describe the steps taken to build and analyze Model 5.

Further note that `Insurance_train` and `Insurance_test` datasets with the ungrouped `CHILDREN` and `REGION` predictors are used to build Model 5. This is because we choose to fully utilize the information provided by `CHILDREN` and `REGION` to achieve the highest predictive accuracy.

**Step 1 - load the library to fit boosted regression trees.**

```
library(gbm)
```

**Step 2 - fit the model on the training data and obtain some summary plots and statistics.**

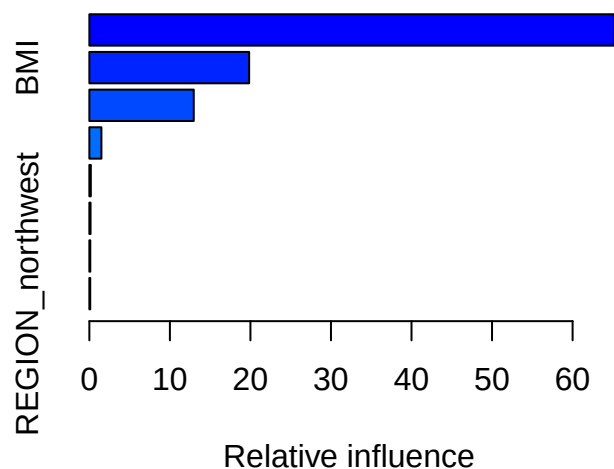
First please note that in the code below the optimal `interaction.depth` and `shrinkage` parameters have been established empirically as well as using some theoretical grounds<sup>10</sup>.

Based on the relative influence plot and the relative influence statistics below, predictors such as `SMOKER_yes`, `BMI` and `AGE` are by far the most important variables. This is further supported by the partial dependence plots below.

```
boosting_model <- gbm(CHARGE ~., data= Insurance_train,
                      distribution = "gaussian", n.trees = 5000,
                      interaction.depth = 7, shrinkage = 0.0008, verbose = F)
summary(boosting_model)
```

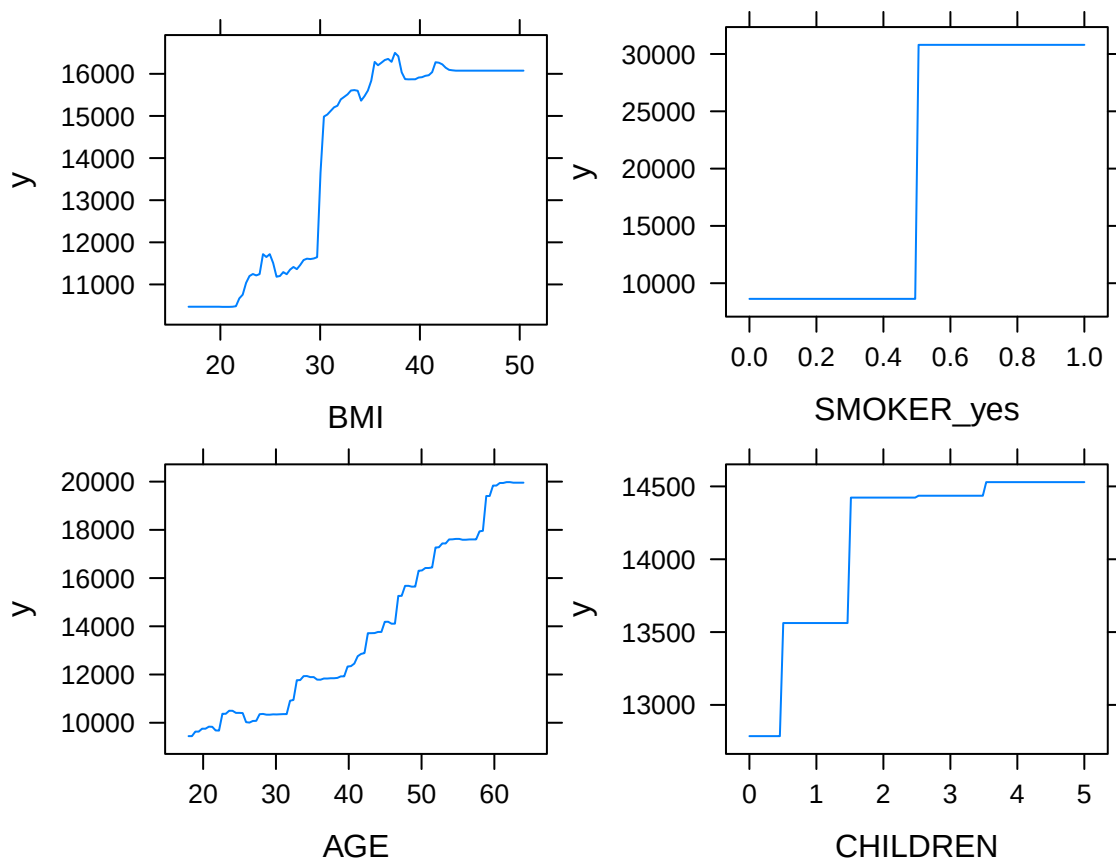
---

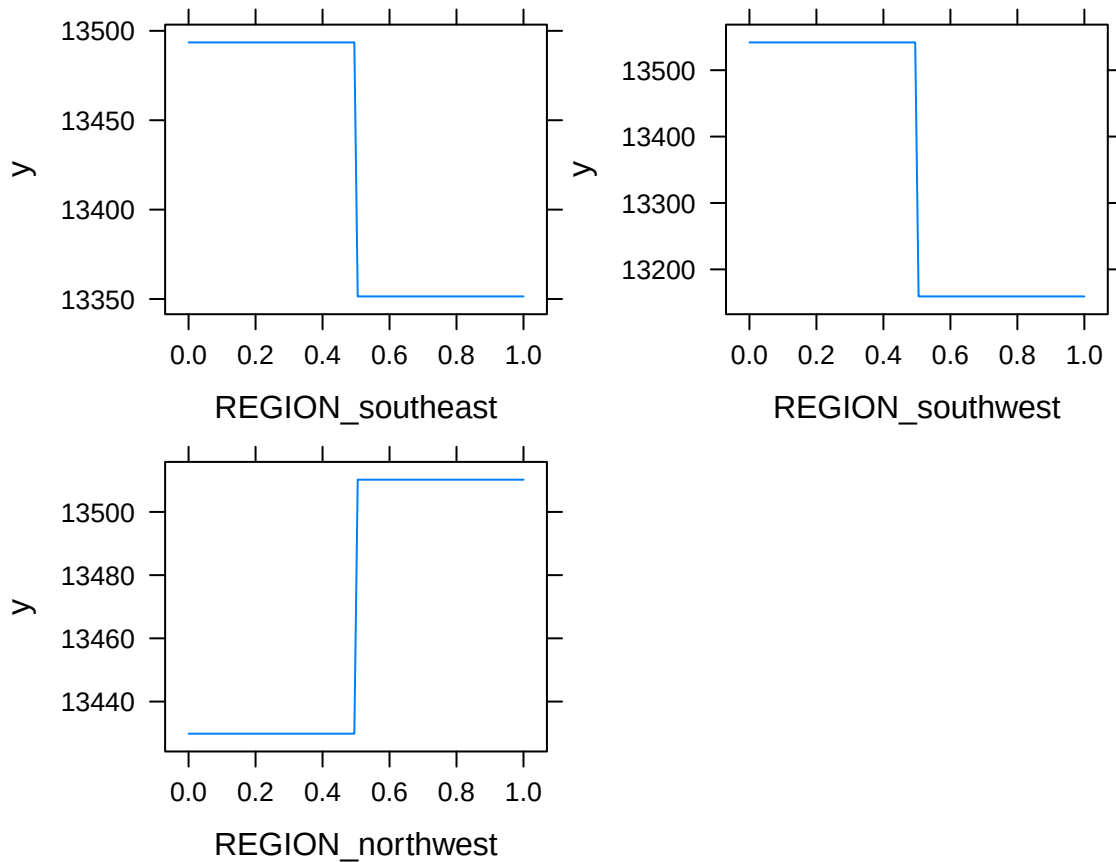
<sup>10</sup>StackExchange, 2015, *How to find optimal values for the tuning parameters in boosting trees*, viewed 2 February 2022, <https://stats.stackexchange.com/questions/103495/how-to-find-optimal-values-for-the-tuning-parameters-in-boosting-trees>



```
##               var    rel.inf
## SMOKER_yes      SMOKER_yes 65.1467770
## BMI              BMI      19.8398363
## AGE              AGE      12.9640444
## CHILDREN         CHILDREN  1.4958721
## REGION_southwest REGION_southwest 0.1806480
## REGION_southeast REGION_southeast 0.1483824
## SEX_male         SEX_male  0.1142969
## REGION_northwest REGION_northwest 0.1101429
```

Partial dependence plots for the predictors:



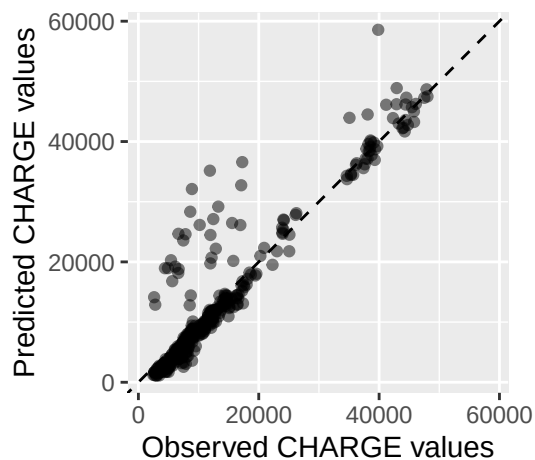


The plots illustrate the marginal effects of **SMOKER\_yes**, **BMI** and **AGE** on **CHARGE** after integrating out other variables. As expected, medical costs billed by health insurance increase with both **BMI** and **AGE**. Moreover, higher bills are associated with people who smoke, have more dependents and live in the north of the US. This goes in line with conclusions made for Model 1.

### Step 3 - assessing prediction accuracy.

Based on the test set we calculate the RMSE for Model 1 and find it to be USD 4,305

Then we plot predicted **CHARGE** values against the observed ones for further illustration:



**CONCLUSION:** we obtained a highly flexible and relatively interpretable Model 5 through boosting regression trees. Most importantly, Model 5 has achieved the highest prediction accuracy compared to Models 1 to 4.

## SUB-SECTION 5.6 - MODELS SUMMARY

We obtained five distinct models using various Machine Learning techniques. The information on the models is summarized below.

Table 8: MODELS SUMMARY

| Model                                | Predictive<br>RMSEAccuracy | Degree of<br>interpretability | Objective                                                                  |
|--------------------------------------|----------------------------|-------------------------------|----------------------------------------------------------------------------|
| Model 1 - Multiple Linear Regression | 5,688 medium               | highest                       | Baseline model                                                             |
| Model 2 - SGD                        | 5,514 medium               | highest                       | Gradient descent based model                                               |
| Model 3 - GAM                        | 5,614 medium               | medium                        | Non-baseline (relatively) interpretable model                              |
| Model 4 - High Dimensional           | 6,450 lowest               | medium                        | (Relatively) high-dimensional model                                        |
| Model 5 - Boosted Regression Trees   | 4,305 highest              | lowest                        | (Relatively) more focused on predictive accuracy without interpretability. |

The Machine Learning methods were selected in such a way that each of the five models is tailored specifically to meet the aim of the study. In our opinion, the most interpretable model was found to be Models 1 and 2, whilst the highest predictive accuracy was achieved by Model 5. Overall, we believe that these models could be used in practice to analyze and predict insurance premiums to help insurance companies stay profitable. We acknowledge that there are limitations associated with our models and we discuss them in the next section.

## SECTION 6 - LIMITATIONS AND IMPROVEMENTS

We realize that each model has its own limitations for which we suggest improvements.

### *MULTIPLE LINEAR REGRESSION MODEL 1*

**Limitation:** Model 1 is built based on the best subsets variables selection method. For the method to be effective we assumed that the number of predictor variables in Model 1 is not so large compared to the number of observations in the data. However, if our assumption was incorrect, best subset may result in overfitting the training data and giving poor predictions for the test data. Relatively high RMSE values were indeed observed in Model 1. Overfitting may further result in high variance for the coefficient estimates, hindering Model 1 interpretability.

**Improvement:** Explore other variable selection methods such as stepwise methods to see if the same subset of predictors can be achieved.

### *STOCHASTIC GRADIENT DESCENT MODEL 2*

**Limitation:** Using Stochastic Gradient Descent to fit Model 2 might be computationally expensive. This is because unlike Gradient Descent, where the parameters are updated on the entire training sample, SGD is updating them on each individual random training sample obtained as part of SGD calculation. Additionally, minimizing an alternative loss function instead of the MSE function might lead to higher predictive



accuracy<sup>11</sup>.

**Improvement:** Explore minimizing an alternative loss function such as quadratic error loss as well as implementing Gradient Descent or Mini-Batch Gradient Descent.

### ***GAM MODEL 3***

**Limitation:** The GAM model obtained is restricted to be additive, meaning that important interactions between the response variables are missed. Put simply, from the EDA it is clear that there exist interactions between **SMOKER** and **AGE**. This, however, is not taken into account by the model, potentially hindering its predictive accuracy.

**Improvement:** Manually add interaction terms to the GAM model, similarly to how it was done in Model 4.

### ***HIGH-DIMENSIONAL MODEL 4***

**Limitation:** Model 4 contains a large number of interaction terms which is likely to create a severe problem of multicollinearity between the variables in this multiple linear regression model. As this reduces the accuracy of the coefficient estimates, we cannot obtain valid associations between each predictor variable and the outcome **CHARGE**.

**Improvement:** Obtain VIF values for Model 4 to see the effects of introducing these interaction terms. If these are alarmingly high, identify only those interaction terms for which there is no problem of multicollinearity.

### ***BOOSTED REGRESSION TREE MODEL 5:***

**Limitation:** Model 5's predictive accuracy could have been impacted by the fact that the boosting procedure used is sensitive to outliers. In other words, given that the values of **CHARGE** are highly positively skewed, outliers may be present in the data. These points could have been removed at the data editing stage of the analysis.

**Improvement:** To see if retaining or removing outliers has a substantial impact on Model 5's predictive power, explore building two models based on the datasets with and without outliers and compare their predictive accuracy.

## **SECTION 7 - FURTHER WORK DIRECTIONS**

In the future our study of insurance premiums could potentially be further extended in the following three directions.

Firstly, in the above study we use machine learning techniques applicable to a continuous outcome variable. In order to be able to apply other machine learning methods the outcome variable **CHARGE** could be converted to a categorical variable. This would allow us to apply other techniques such nearest neighbours, logistic regression, support vector machines and so on.

Secondly, the five models above were based on only six predictors available in the Kaggle dataset. More thorough models could potentially be built by considering other factors which may affect individual medical costs billed by health insurance such as income, job occupation, medical history, profession and so on.

Thirdly, our study was focused on the US only. We could potentially explore individual medical costs billed by medical insurance in other countries to account for differences between various countries such as social, geographical, ethnical and cultural.

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<sup>11</sup>Donges, N., 2021, *Gradient Descent: An Introduction to 1 of Machine Learning's Most Popular Algorithms*, viewed 2 February 2022, <https://builtin.com/data-science/gradient-descent>

## SECTION 8 - CONCLUSION

In our study we attempted to predict and explore social and demographic factors that might be associated with the medical costs billed by health insurance companies and hence health insurance premiums to advise actuaries on pricing the health insurance policies. We used publicly available data and analysed it with the help of the five statistical models built using multiple machine learning techniques.

We constructed five models each tailored to the aim of the study. We believe that Models 1 and 2 are the most interpretable ones, whilst Model 5 achieved the highest predictive accuracy. The rest of the models achieve interpretability and predictive accuracy goals to various acceptable extents.

Overall, taking into account limitations of the study and the need for further improvement in some areas, we conclude that being older and having higher BMI is associated with higher health insurance premiums. Moreover, smoking, having more dependents and living in the North of the US are further factors increasing insurance premiums. Therefore, we believe, our study might be of help for professionals who determine the health insurance premiums to ensure profitability of the insurance companies.

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